Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Module 2
Linear Transformation and Matrix Decomposition

Module 3
Probability Foundations for Statistical Learning

Lesson 1: Sum Rule, Product Rule, and Bayes' Theorem

- Sum Rule, Product Rule, and Bayes' Theorem — Reading • 15 min
- Sum Rule — Reading • 4 min
- Product Rule (Chain Rule) — Reading • 2 min
- Bayes' Theorem — Reading • 43 min

Lesson 2: Gaussian Distribution

- Univariate Gaussian Distribution — Reading • 4 min
- Gaussian Distribution: Foundations and Applications — Reading • 15 min
- Multivariate Gaussian Distribution — Reading • 10 min
- Conditional and Marginal Multivariate Gaussian Distributions — Reading • 45 min
- Product of Gaussian Densities — Reading • 26 min

Lesson 3: Inference Techniques

- Bayesian Inference — Reading • 1h 10m
- **Latent-Variable Models** — Reading • 50 min
- Check Your Knowledge: Inference Techniques — Graded Assignment • 10 min

Module 4
Introduction to Model Evaluation and Optimization

Latent-Variable Models

0:31 / 5:02 2x

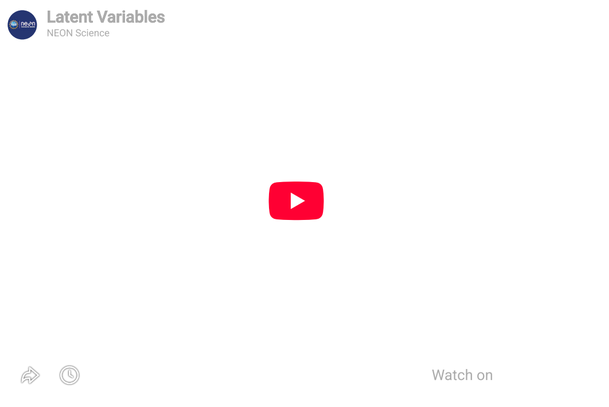
Latent variables are hidden variables in a model that you can't directly see or measure. Even though they aren't explicitly part of the model's parameters, they can describe how the data is generated. These variables help make models easier to interpret and reduce complexity by requiring fewer parameters.

Examples of latent variables include:

- Missing data (values you don't have but might need)
- Data measured with errors (like biased or random mistakes in measurements)
- Proxy measures (things you measure that stand in for something else)

Ignoring these hidden variables can lead to mistakes. For example, relying on incomplete or flawed observations can result in models making overly confident but incorrect predictions.

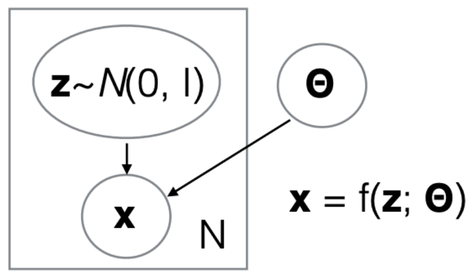
Watch this next video where you'll learn how these hidden factors influence data generation and model behavior. Through clear explanations and examples, this video explores why understanding and accounting for latent variables is crucial for building accurate and reliable machine learning models.



In a latent variable model:

- We start with an observed output, x, which we assume to be a continuous vector.
- There's a hidden variable, z, that is generated from a Gaussian distribution (a normal distribution).
- A function, f, maps the hidden variable z to the observed output $_x$. This function is controlled by parameters, Θ, and is typically represented by a neural network.

The process can be summarized as: x is produced by applying the function $f(z; \Theta)$, where z follows a Gaussian distribution. The diagram shows this relationship visually.

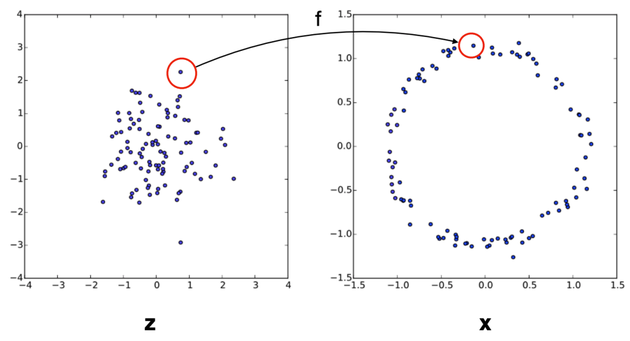


Neubig, G. (2020). Neural networks for NLP: Latent variable models. Retrieved from Phontron.

Complex distribution (of (x)) is characterized by a Gaussian distribution over the latent variable. A deterministic mapping function maps the variable in the latent space ((z)) to the output ((x)).

Why it is interesting: You can take the Gaussian space and arbitrarily transform it to obtain any other complex distribution, such as mapping to a circle in (x) or any other natural image in the world.

The image below illustrates an example of how a latent variable, z, is transformed into an observed variable, x, through a function, f.



Neubig, G. (2020). Neural networks for NLP: Latent variable models. Retrieved from Phontron.

The plot above shows a set of points sampled from a 2D latent space, z, which appears as a cluster of points distributed around the origin. The plot on the right displays the corresponding points in the observed space, x, after applying the transformation function, f. The red circles highlight how a single point in the latent space maps to a specific point in the observed space, demonstrating the role of the function in shaping the data structure (e.g., from a cluster to a circular distribution). This transformation showcases how latent variable models encode hidden structure and patterns into observed data.

Building on this concept, as introduced earlier, the latent variable model illustrates how data is generated using a latent variable (z) sampled from a prior distribution $(p(z))$ and transformed into an observed variable (x) through a function (f) parameterized by theta (θ). This framework is now extended to explain how Variational Auto-Encoders (VAEs) apply these probabilistic principles.

Benefits of Latent Variables:

- Latent variables make the model structure simpler and richer, facilitating tasks like dimensionality reduction (PCA), density estimation (Gaussian mixture models), and time-series modeling (Hidden Markov Models).
- Learning in latent-variable models can be effectively done using the Expectation-Maximization (EM) algorithm.

Posterior on the Latent Variables

$$p(\boldsymbol{z} \mid \mathcal{X}) = \frac{p(\mathcal{X} \mid \boldsymbol{z}) p(\boldsymbol{z})}{p(\mathcal{X})}, \quad p(\mathcal{X} \mid \boldsymbol{z}) = \int p(\mathcal{X} \mid \boldsymbol{z}, \boldsymbol{\theta}) p(\boldsymbol{\theta}) \mathrm{d}\boldsymbol{\theta}$$

A quantity that is easier to compute is the posterior distribution on the latent variables, but conditioned on the model parameters

$$p(\boldsymbol{z} \mid \mathcal{X}, \boldsymbol{\theta}) = \frac{p(\mathcal{X} \mid \boldsymbol{z}, \boldsymbol{\theta}) p(\boldsymbol{z})}{p(\mathcal{X} \mid \boldsymbol{\theta})}$$

Modern latent variable modeling comprises a large collection of useful models and strategies for mental health research. Indeed, one may argue the notion of the latent variable is perhaps the single most important concept exported from the psychological sciences to the statistical sciences.

When exploring latent variable modeling in greater depth, this resource, Latent Variable Modeling (36 minutes), provides a comprehensive overview. It covers foundational concepts, practical applications, and statistical techniques, making it an excellent guide for understanding how to model hidden structures in data.

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